

FanSphere AI

Football Audience Intelligence

Technical Report • Version 1.1

Prashant Upadhyay • June 2026

Separating engagement volume from engagement intensity across 907,158 fan comments and 10 La Liga fixtures, then tuning the model with an autonomous experiment loop.

github.com/Prashant-upadhyay-Projects/FanSphere-AI

1 Executive Summary

FanSphere AI measures how football matches move fan communities. It joins on-pitch event data from StatsBomb with subreddit conversation from Reddit, scores sentiment at the comment level, segments authors into behavioural cohorts, and produces a composite engagement score per fixture that separates raw conversation volume from emotional intensity.

The headline result: once the scoring blend is weighted toward the fan signal, El Clasico rises to the top of the combined engagement ranking. Crucially, volume and rank are not the same thing. The single most-discussed fixture in the dataset (17,089 comments) ranks only fifth, because it was a low-scoring match. Volume, fan affect, and on-pitch excitement behave as three distinct axes, and the divergence between them is the product's core insight.

Version 1.1 replaced two unvalidated modelling defaults with results from an autonomous experiment loop. Author clustering moved from KMeans k=3 with standard scaling to k=4 with robust scaling, lifting the silhouette score from 0.40 to 0.64 and surfacing a distinct power-user cohort. The engagement blend moved from an even 0.5/0.5 split to a fan-biased 0.35/0.65, which lifted El Clasico from 4th to 1st and improved how cleanly rivalry fixtures separate from the rest (AUC 0.50 to 0.81).

2 The Problem

Commercial and content teams at top clubs consume the same primitives: who is engaging, when, how strongly, and against which fixture. FanSphere AI implements those primitives on fully open data, doubling as a reference architecture for fan behavioural intelligence and a research probe into whether on-pitch events drive online fan affect or merely correlate with it.

The non-obvious move is what happens after both signals exist. Scoring each fixture on an on-pitch axis and a fan axis and ranking them is the obvious step. The insight is that the two axes can diverge, and that divergence is itself the finding worth surfacing.

3 System Architecture

A linear Python pipeline feeds a single DuckDB file, which an Evidence.dev dashboard reads directly. Every stage is auditable and reproducible from open inputs.

Stage	Module	Responsibility
Ingest	load_statsbomb.py	Fixtures, scorelines, rivalry tags, xG
Ingest	load_reddit_archive.py	Pushshift JSONL to parquet (907K comments)
Link	link_comments_to_matches.py	Confidence-scored comment-to-fixture joins (threshold 0.40)
Score	sentiment.py	VADER compound sentiment per comment
Segment	generate_fan_segments.py	KMeans k=4 + RobustScaler author cohorts
Aggregate	engagement.py	Composite score: 0.35 football + 0.65 fan
Serve	Evidence.dev + DuckDB	6-page interactive dashboard, 9 tables

4 Methodology

Data.

StatsBomb open data provides 10 La Liga 2020/21 fixtures (Barcelona-centric) with 53 goals. Reddit Pushshift archives provide 907,158 comments from r/barca and r/realmadrid. Of those, 93,298 link to a fixture.

Linking.

Each comment receives a confidence score for which fixture it belongs to, composed from soft signals: both teams mentioned (0.60), one team (0.35), rivalry keyword (0.25), home subreddit (0.25), generic football subreddit (0.10), within a 48-hour window of kickoff. Links below 0.40 are dropped. The approach is rule-based rather than a trained classifier so that every link is auditable, with no hand-labelling required.

Sentiment.

VADER scores each comment to a compound value in the range minus one to plus one. It is rule-based, deterministic, fast on a million comments, and fully auditable, which matters more here than the marginal accuracy gain of a transformer. The interface is abstracted so a transformer can be dropped in later.

Segmentation.

Authors with at least three comments are clustered on comment frequency, sentiment volatility, and total upvotes received. Version 1.1 uses KMeans k=4 with RobustScaler. Clusters are relabelled by mean engagement so the cohort names are stable across runs.

Engagement scoring.

The fan-side score blends log-scaled volume (0.45), absolute mean sentiment (0.20), sentiment volatility (0.25), and cross-subreddit reach (0.10), each min-max normalised. The combined score is 0.35 times the normalised on-pitch score plus 0.65 times the fan-side score.

5 Version 1.1: Tuning via AutoResearch

Two numbers in version 1.0 were unvalidated defaults: k=3 clustering and the even 0.5/0.5 blend. Version 1.1 tested them with a Karpathy-style ratchet loop. The loop proposes one change, runs a frozen evaluator that returns a single score, keeps the change only if it strictly beats the current best, otherwise reverts it from a snapshot, and logs every attempt.

A label-grounded metric.

The decision the whole loop hinges on is what better means. Clustering was scored on 0.6 times silhouette plus 0.4 times a bootstrap stability measure. Engagement was scored on how cleanly the per-fixture score separates the two rivalry matches from the other eight, using the `is_rivalry` and `total_goals` labels already present in the data. No hand-labelling was required, and the metric rewards a real quality rather than a property that can be trivially gamed.

Guardrails.

Because the loop ran on a flat-rate plan, a usage guardrail paused it at 60 percent of the rolling 5-hour usage window, backed by a hard iteration cap and a graduation kill-switch. Nine experiments ran in total: six kept, two discarded, one reverted. Promotion to production was manual and reviewed; the loop never edited production code directly.

Results.

Decision	v1.0	v1.1	Result
Author clustering	k=3, StandardScaler	k=4, RobustScaler	silhouette 0.40 to 0.64; new Ultra cohort
Engagement blend	0.5 / 0.5	0.35 / 0.65	El Clasico 4th to 1st; rivalry AUC 0.50 to 0.81
Sentiment model	VADER, mean	unchanged	all 6 combos tested; hit a ceiling, not the bottleneck

Table 1. The two changes that shipped, and the one deliberate null result.

6 Key Findings

Rank	Fixture	Goals	Combined	Fan
1	El Clasico: Barcelona 1-3 Real Madrid	4	0.598	0.681
2	Real Sociedad 1-6 Barcelona	7	0.597	0.380
3	Levante 3-3 Barcelona	6	0.558	0.500
4	Barcelona 5-2 Real Betis	7	0.527	0.272
5	El Clasico: Real Madrid 2-1 Barcelona	3	0.435	0.609

Table 2. Top five fixtures by combined engagement (v1.1).

The October El Clasico ranks first: it carries the highest fan-affect score in the dataset (0.681), and the fan-biased blend rewards that. The April El Clasico drew the most comments of any fixture (17,089) yet ranks fifth, because a 3-goal match scores low on the on-pitch axis and the combined score still respects both signals. This is the volume-versus-intensity divergence the project was built to surface.

Fan segmentation produced four cohorts on an engagement-intensity ladder: Casual (3,150 authors), Tactical (359), Highly Engaged (100), and a small Ultra tier (19) that posts roughly 500 comments each. The tiers differ in volume and influence rather than in how they argue.

7 Limitations

- Small sample. 10 fixtures with only 2 rivalry matches makes the rivalry metric coarse and the claims descriptive of this dataset, not population-level.
- Single-team bias. The corpus skews Barcelona-side; 8 of 10 fixtures involve Barcelona.
- VADER cannot read sarcasm or language switches (Spanish, Catalan), which a domain-tuned transformer would handle better.
- Linking confidence is fuzzy. Most comments sit at the 0.60 floor (subreddit plus time window), so the directional signal is defensible but individual links are not guaranteed.

8 Future Work

- Replace VADER with a football-domain transformer. v1.1 confirmed the lexicon is at its ceiling, so the next gain must come from a better model.
- Expand across multiple seasons and competitions to make statistical claims population-level.
- Live ingestion via the Reddit API with fixture-aware temporal windowing.
- Probabilistic match-state modelling driven by xG timelines.
- Cohort stability analysis across consecutive seasons to measure fan drift.

*Full source, the AutoResearch framework, and the interactive dashboard:
github.com/Prashant-upadhyay-Projects/FanSphere-AI*